Exercise Sheet 1

May 16, 2026

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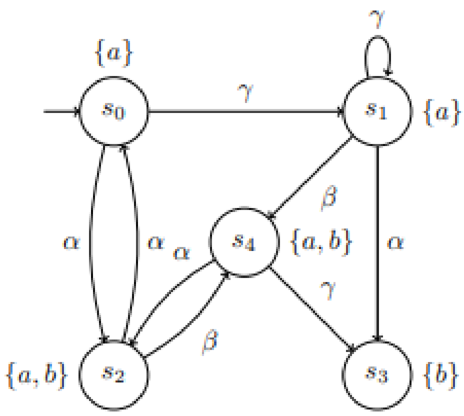
1 Exercise 1

We call a transition system $TS = (S, Act, \rightarrow, S_0, AP, L)$

- action-deterministic if $|S_0| \leq 1$ and $|Post(s, \alpha)| \leq 1$ for all $s \in S$ and $\alpha \in Act$, and
- AP-deterministic if $|S_0| \leq 1$ and $|Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1$ for all $s \in S$ and $A \in 2^{AP}$,

where $Post(s, \alpha) = \{s' \in S \mid \exists\, (s, \alpha, s') \in \rightarrow\}$ and
$Post(s) = \cup_{\alpha \in Act} Post(s, \alpha)$

Let the transition system TS_1 be as follows.



(a) Give the formal definition of TS_1

(b) Specify a finite and an infinite execution of TS_1

(c) Decide whether TS_1 is (i) AP-deterministic, and/or (ii) action-deterministic. Justify your answer.

Solution

(a) The formal definition of TS_1:

- State Space: $S = \{s_0, s_1, s_2, s_3, s_4\}$
- Action: $Act = \{\alpha, \beta, \gamma\}$
- Transition relation:

$$\rightarrow = \{(s_0, \gamma, s_1),\ (s_0, \alpha, s_2),\ (s_1, \gamma, s_1),\ (s_1, \beta, s_4),\ (s_1, \alpha, s_3),\ (s_2, \alpha, s_0),\\ (s_2, \beta, s_4),\ (s_4, \alpha, s_2),\ (s_4, \gamma, s_3)\}$$

- Initial State: $S_0 = s_0$
- Atomic Propositions: $AP = \{a, b\}$
- Labeling Function:

$$L(s) = \begin{cases} \{a\} & \text{if } s \in \{s_0, s_1\} \\ \{a, b\} & \text{if } s \in \{s_2, s_4\} \\ \{b\} & \text{if } s = s_3 \end{cases}$$

(b) A finite execution of TS_1:

$$s_0 \xrightarrow{\alpha} s_2 \xrightarrow{\beta} s_4 \xrightarrow{\gamma} s_3$$

An infinite execution of TS_1:

$$(s_0 \xrightarrow{\alpha} s_2 \xrightarrow{\alpha} s_0)^{\omega}$$

(c) Based on what has been given, we have:

- $\forall s \in S, \forall \alpha \in Act$, if $(|S_0| \leq 1) \wedge (|Post(s, \alpha)| \leq 1) \implies$ Action-deterministic (1)
- $\forall s \in S, A \in 2^{AP}$, if $(|S_0| \leq 1) \wedge (|Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1)$ $\implies$ AP-deterministic (2)

Because $|S_0| = 1 \implies |S_0| \leq 1 = True$ (3)

- Check TS_1 is AP-deterministic

  2^{AP} is the set of all subsets of AP

  $$2^{AP} = \{\phi, \{a\}, \{b\}, \{a, b\}\}$$

  From (2), (3) $\implies$

  For TS_1 is AP-deterministic, $|Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1$ must be True

  - With $s = s_0$:
    $Post(s_0) = \{s_1, s_2\}$
    - With $A = \phi$
      $\{s' \in S \mid L(s') = A\} = \{\}$
      $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
    - With $A = \{a\}$
      $\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$
      $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_1\}| = 1 \leq 1$ (satisfied)
    - With $A = \{b\}$
      $\{s' \in S \mid L(s') = A\} = \{s_3\}$
      $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
    - With $A = \{a, b\}$
      $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
      $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_2\}| = 1 \leq 1$ (satisfied)
  - With $s = s_1$
    $Post(s_1) = \{s_3, s_4\}$
    - With $A = \phi$
      $\{s' \in S \mid L(s') = A\} = \{\}$
      $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
    - With $A = \{a\}$
      $\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$
      $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)

- With $A = \{b\}$
  $\{s' \in S \mid L(s') = A\} = \{s_3\}$
  $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_3\}| = 1 \leq 1$ (satisfied)
- With $A = \{a, b\}$
  $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
  $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_4\}| = 1 \leq 1$ (satisfied)

– With $s = s_2$
  $Post(s_2) = \{s_0, s_4\}$

  - With $A = \phi$
    $\{s' \in S \mid L(s') = A\} = \{\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
  - With $A = \{a\}$
    $\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_0\}| = 1 \leq 1$ (satisfied)
  - With $A = \{b\}$
    $\{s' \in S \mid L(s') = A\} = \{s_3\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
  - With $A = \{a, b\}$
    $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_4\}| = 1 \leq 1$ (satisfied)

– With $s = s_3$
  $Post(s_3) = \{\}$

  - With $A = \phi$
    $\{s' \in S \mid L(s') = A\} = \{\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
  - With $A = \{a\}$
    $\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
  - With $A = \{b\}$
    $\{s' \in S \mid L(s') = A\} = \{s_3\}$

$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)

- With $A = \{a, b\}$
  $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
  $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)

– With $s = s_4$
  $Post(s_4) = \{s_2, s_3\}$

  - With $A = \phi$
    $\{s' \in S \mid L(s') = A\} = \{\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
  - With $A = \{a\}$
    $\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
  - With $A = \{b\}$
    $\{s' \in S \mid L(s') = A\} = \{s_3\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_3\}| = 1 \leq 1$ (satisfied)
  - With $A = \{a, b\}$
    $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
    $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_2\}| = 1 \leq 1$ (satisfied)

$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1 =$ True (4)
From (3), (4) $\implies TS_1$ is AP-deterministic

- Check TS_1 is Action-deterministic
  From (1) and (3) $\implies$ For TS_1 is action-deterministic, $|Post(s, \alpha)| \leq 1$ must be True

  – Case $s = s_0$:
    $Post(s_0, \gamma) = \{s_1\}$, $Post(s_0, \alpha) = \{s_2\}$ (satisfied)
  – Case $s = s_1$:
    $Post(s_1, \beta) = \{s_4\}$, $Post(s_1, \alpha) = \{s_3\}$ (satisfied)
  – Case $s = s_2$:
    $Post(s_2, \beta) = \{s_4\}$, $Post(s_2, \alpha) = \{s_0\}$ (satisfied)

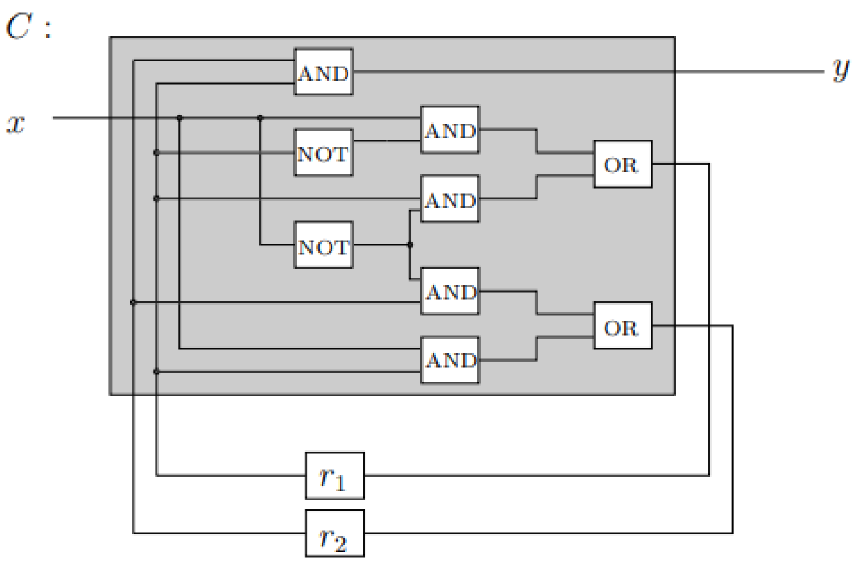
– Case $s = s_3$: (satisfied because there are no way to reach to any state)
– Case $s = s_4$:
$Post(s_4, \alpha) = \{s_2\}$, $Post(s_4, \gamma) = \{s_3\}$ (satisfied)

$\implies |Post(s, \alpha)| \leq 1 = \text{True}$ (5)

From (3), (5) $\implies TS_1$ is action-deterministic

2 Exercise 2

Consider the following sequential hardware circuit in C.



(a) Give the function λ_y, δ_{r1} and δ_{r2} that govern the changes to the output and register bits depending on the input bits.

(b) Formally specify the state space of the transition system TS for the circuit C. Justify your answer.

(c) Draw the transition system TS for the circuit C using the set of labels $AP = \{x, r_1, r_2, y\}$ assuming that initially the values of the registers are 0. Make sure that **all formal components** of the transition system can be uniquely identified from your picture.

(d) Determine the set $Reach(TS)$.

Solution

3 Exercise 3*

Consider the following mutual exclusion algorithm for two processes P_0 and P_1.
The pseudo code of the algorithm for process P_i is given as

$$
\begin{array}{rl}
 & \text{int k := 0;} \\
 & \text{b := [true, true];} \\
\ell_0 & \left\{ \begin{array}{l} \text{while (true) do} \\ \quad b[i] := \textit{false}; \end{array} \right. \\
\ell_1 & \left\{ \quad \text{while (k != i) do} \right. \\
\ell_2 & \left\{ \begin{array}{l} \qquad \text{while (not b[1-i]) do} \\ \qquad\quad \text{k} := i; \\ \qquad \text{end} \end{array} \right. \\
 & \quad\ \text{end} \\
\ell_3 & \left\{ \quad \text{critical\_section;} \right. \\
\ell_4 & \left\{ \quad \text{b}[i] := \text{true;} \right. \\
 & \text{end}
\end{array}
$$

where b is an array of two Boolean values which are initially true. k is a variable which is either 0 or 1, and initially 0.

(a) Give the program graph representation for a single process. Use the locations $\ell_0, \ldots \ell_4$ corresponding to the indicated program fragments.

(b) Give the reachable part of the transition system of $P_0|||P_1$, i.e., $\mathrm{TS}(P_0|||P_1)$, over the state space $\langle \ell_i, \ell_j, k, b[0], b[1] \rangle$.
Hint: If you wish, you may treat the states $\langle \ell_i, \ell_j, 0, b[0], b[1] \rangle$ and $\langle \ell_j, \ell_i, 1, b[1], b[0] \rangle$ to be equivalent. In other words, whenever you encounter a transition to a state with $k = 1$, you may swap the locations of the processes P_0 and P_1, swap the values of $b[0]$ and $b[1]$, set k to 0, and make the transition target this "new" state instead. This operation considerably reduces the size of the resulting state space.

(c) Check whether the algorithm ensures mutual exclusion, i.e., both processes are never in their critical section at the same time. Justify your answer.

(d) Does the algorithm guarantee that every process eventually enters its critical section? Justify your answer.